

| Document Control |                      |
|------------------|----------------------|
| Document Name    | Safe Reach - WIS PRD |
| Version No       | 1.0                  |
| Classification   | Internal             |
| Date             | 03/06/2025           |

| Version Control |                  |             |               |                  |
|-----------------|------------------|-------------|---------------|------------------|
| VerNo           | Author           | Approved by | Approved Date | Description      |
| 1.0             | Vaishnavi Rag... |             |               | Initial Document |

## SAFE REACH - WIS PRD

### Problem Statement:

Organizations require a robust mechanism to ensure the safety and compliance of employees, particularly those leaving office premises after designated late hours (e.g., 8 PM). Traditional methods of confirmation are insufficient, and the existing system lacks the capabilities to:

1. Consistently record and confirm an employee's safe departure and arrival, especially when they may or may not have an office booking or when auto-checkout is enabled.
2. Capture necessary additional safety-related data (e.g., emergency contacts, mode of transport, drop location) at the point of departure, which is crucial for compliance and emergency response, especially after certain hours.

3. Trigger safety workflows (like Safe Reach calls/notifications) based on specific criteria such as gender, which is a compliance requirement for some companies.
4. Allow employees to define their Estimated Time of Arrival (ETA), leading to untimely notifications that reduce the effectiveness of the safe reach verification process.
5. Accurately track and report all relevant data, including dynamically updated contact numbers and custom safety fields, for audit and compliance purposes, as current reports are often incomplete or hardcoded.
6. Provide a centralized, accessible, and real-time dashboard for security teams to monitor and manage safe reach statuses effectively without relying on disparate systems.

### **Personas involved:**

1. Employee: An individual working at the organization, potentially leaving the office after designated hours.
2. Office Admin: Responsible for managing office-specific configurations, monitoring employee safety, and receiving relevant alerts.
3. Security Guard / Security Team Personnel: Responsible for monitoring the Safe Reach dashboard, performing manual verifications, and ensuring employee safety.

### **Designs:**

[Employee Flow \(VMS\)](#)

[Safe Reach \(Custom Fields\)](#)

[Safe Reach Dashboard](#)

### **Solution (Overview)**

The "Safe Reach" feature in WorkInSync (WIS) provides a robust safety mechanism for employees leaving office premises, especially after designated late hours. Unlike the existing ETS Safe Reach which involves company cabs, WIS Safe Reach operates independently of company-provided transport, allowing employees to use their preferred mode of transport while ensuring their safety is still monitored by office admins and security teams.

The core solution involves:

1. **VMS Kiosk Integration:** Employees initiate Safe Reach checkout via the VMS self check-in kiosk.

2. **Configurable Data Capture:** Dynamic forms capture essential safety information (e.g., emergency contact, mode of transport, drop location) based on client's needs.
3. **Flexible Notification Triggers:** Safe Reach notifications (mobile, IVR) are triggered based on configurable ETAs provided by the employee or system defaults, with gender-based triggering options.
4. **Automated Verification & Escalation:** Automated mobile notifications and IVR calls verify safe arrival, with escalation to manual verification by security personnel if automated attempts fail.
5. **Enhanced Communication:** Clear, timely, and office-specific email notifications for failed verifications.
6. **Comprehensive Reporting & Dashboard:** A real-time, configurable dashboard and detailed reports provide complete visibility into Safe Reach statuses and captured data for audit and compliance.
7. **Office-Level Customization:** Key safety parameters (like trigger times and email recipients) are configurable at the office level, overriding BUID defaults for localized compliance.
8. **NDA Signature Capture:** Supports signature capture for NDA consent during checkout for stricter compliance requirements.

## Use Cases

### 5.1 The Employee Persona

#### Use Case 1.0: Initiating Safe Reach Checkout from VMS Kiosk

1. Description: An employee leaves the office and initiates the Safe Reach workflow through the VMS self check-in kiosk to confirm their departure and trigger the Safe Reach process.
2. Flow:
  - a. Employee navigates to the VMS self check-in kiosk.
  - b. Selects "I am an Employee" option.
  - c. Choose the "Checkout" option.
  - d. Selects either "Checkout via Email" or "Checkout via Phone" and enters the respective ID or phone number.
  - e. Enter OTP for verification.

- f. The system validates the entered email/phone number against the employee database. If the entered details do not match the employee's DB record, the Safe Reach flow will not trigger.
  - g. The visibility of Safe Reach specific input fields depends on the time of day and system settings (refer to **Key Configurations/Properties** for details on SafeReachVmsTimeInMin)
  - h. If a visitor tries to check out, Safe Reach specific input fields will not be visible, regardless of time.
3. System Behavior on Check-out
- a. Employee has a booking for the day whose check-out is before 8 PM:
    - i. If auto-checkout is set up and the employee was already checked out: The relevant information needs to be shared with the Safe Reach module upon manual checkout.
    - ii. If auto-checkout is not set up and the booking was not checked out: The employee's booking will be checked out, and relevant information sent to the Safe Reach module.
  - b. Employee has a booking for the day whose check-out is post 8 PM:
    - i. The employee's booking will be checked out, and relevant information sent to the Safe Reach module.
  - c. Employee does not have a booking for the day:
    - i. Relevant information will be sent to the Safe Reach module.
  - d. Employee has both visitor and employee records for that tenant:
    - i. Only the employee flow is acted upon based on the above conditions (office booking existence or not). The visitor booking is ignored.
4. Success Message: Once successfully checked-out, the message appears on the success screen: "Safe reach has been initiated. You will receive a call in [Selected ETA or 90] mins to confirm if you have reached your destination. Please answer the call promptly for your safety."
- a. This message is configurable.

#### Use Case 1.1: Providing Additional Information for Safe Reach

1. Problem: Compliance requires additional data capture from employees checking out after a certain time of day.

2. Description: When an employee checks out after a specific cutoff time, they are prompted to provide additional safety-related information through a form with customizable fields.
3. Form Fields: The input fields for this flow are configurable (refer to **Key Configurations/Properties** for details on SafeReachInputFields):
  - a. Employee Email ID: Prefilled from employee DB, not editable.
  - b. Employee Name: Prefilled from employee DB, not editable.
  - c. Mobile Number: Prefilled from employee DB, editable (info is not modified in the employee profile).
  - d. Employee ID: Prefilled from employee DB, not editable.
  - e. Emergency Contact No: Entered by the employee.
  - f. Mode of Transport: Entered by the employee, with configurable dropdown values including an "Others" option with a free text box (e.g., Personal bike, Car, Public transport).
  - g. Vehicle Details: Entered by the employee.
  - h. Drop Location: Entered by the employee.
  - i. Escort Required: Select 'Yes' or 'No'. If 'Yes' is selected and a non-personal mode of transport is chosen, a text box appears for "Escort Guard Name" and options for "Office Gate" or "Home Location".

#### Use Case 1.2: Gender-Based Safe Reach Trigger

1. Problem: Certain company policies require Safe Reach to be triggered only for female employees, and fields should only be shown to them.
2. Description: The Safe Reach workflow can be configured to activate only for female employees based on their gender in the employee profile. Field visibility is also tied to this setting.
3. Behavior: The system's behavior regarding showing Safe Reach fields on VMS and triggering Safe Reach depends on specific properties (refer to Key Configurations/Properties for triggerSafeReachForFemaleOnly and enableSafeReachForBookingTypes).
4. Edge Case: If gender is missing or not set up for an employee, Safe Reach will not be triggered for that employee.

### Use Case 1.3: Selecting Estimated Time of Arrival (ETA) for Safe Reach Notification

1. Problem: Fixed 90-minute default notification time is not optimal for varying employee ETAs.
2. Description: Employees can select their Estimated Time of Arrival (ETA) during checkout, and Safe Reach notifications will be triggered based on this selection.
3. Flow:
  - a. A dropdown field, "ETA to Reach Destination," is present in the Safe Reach checkout kiosk flow.
  - b. The dropdown values are configurable (refer to Key Configurations/Properties for safeReachETAOptions).
  - c. The employee selects their estimated time to reach the destination.
  - d. Upon completing checkout, the success message updates dynamically to reflect the selected ETA.
4. Notification Trigger: The Safe Reach notification (mobile notification, IVR) will be triggered based on the user's selected ETA instead of any default time.

### Use Case 1.4: Providing NDA Consent with Signature

1. Problem: Compliance requires NDA consent via signature for stricter regulations.
2. Description: During Safe Reach checkout, employees can provide consent either via a checkbox or a digital signature, depending on the system's configuration.
3. Consent Mechanism: The mechanism for NDA consent (checkbox or digital signature) is property-driven (refer to Key Configurations/Properties for enableSignatureForConsentSafeReach).
4. Content: The NDA form content and checkbox/signature prompt will be present for the Safe Reach module, with content provided by the client's team.

## 5.2 Office Admin / Security Team Persona

### Use Case 2.0: Monitoring Safe Reach Status from WIS Dashboard

1. Problem: Column irrelevant for WIS use case and lack of refresh functionality hinder effective monitoring.
2. Description: Office admins and security personnel can access a centralized Safe Reach dashboard in WIS, which provides real-time status updates and is highly adaptable.
3. Access: The Safe Reach dashboard is accessible from the WIS side navigation.
4. Column Customization: The dashboard columns on the WIS side are made specific to our use case. Unnecessary columns which were relevant for the ETS side have been removed.
5. Refresh Button:
  - a. A refresh button is added to the security dashboard to update data without reloading the entire page.
  - b. Interaction with Filters: The refresh button retains all selected filters (e.g., status filters like "Failed," "In Progress," "Automated Success," Date Picker, Employee Search Bar, BUID filter).
  - c. It fetches new Safe Reach status updates based on real-time data for the currently selected filters.
  - d. If no data matches filters after refresh, a "No data matching the filter" message is shown.

### Use Case 2.1: Receiving Failed Verification Notifications

1. Problem: Generic subject lines, delayed notifications, and irrelevant CC lists create confusion and hinder timely intervention.
2. Description: Office admins receive clear, timely, and office-specific email alerts for failed Safe Reach verifications.
3. Email Subject Line for Failed Verification:
  - a. Subject: Alert: Unconfirmed Safe Arrival of {Employee Name} ({Employee ID}) - {Office Location Name}. Example: Alert: Unconfirmed Safe Arrival of

Abhijit Patra (123232) - Bangalore HQ.

4. CC List Enhancement:

- a. Reporting Manager is removed from the CC list for WIS clients (ETS clients remain unaffected).
- b. Email recipients for failed verifications are configurable at the office level, with a fallback to a BUID-level setting if not specified (refer to Key Configurations/Properties for safeReachCcList).

5. Timely Notifications:

- a. An additional job will be introduced to send failed verification emails at multiple configurable intervals throughout the day (e.g., every 30 minutes), instead of a single run at 8 AM the next day.

Use Case 2.2: Generating Safe Reach Reports

- 1. Problem: WIS Safe Reach Reports currently display hardcoded columns and do not support custom fields captured during checkout.
- 2. Description: The WIS Safe Reach Report will include all standard fields and any custom fields configured for the BUID, providing comprehensive data for audit and compliance.
- 3. Report Name: WIS Safe Reach Report.
  - a. Custom Field Support: Custom fields configured for the BUID will be displayed in the reports, appearing after the "ETA to Destination" field, in the order they are configured. If a custom field value is null, "NA" will be displayed.
- 4. Report Columns (Summary & Detail View):
  - a. Emp Id
  - b. Employee Email
  - c. Emp Name
  - d. Gender
  - e. Emp Contact
  - f. Safe Reach Contact (The phone number the employee used during Safe Reach checkout)
  - g. ETA to Destination



- h. Time of Leaving Office
  - i. Verification Status (e.g., "Verification Failed", "In Progress", "Successful")
  - j. Verification Type (e.g., "Mobile Notification", "IVR", "Manual")
  - k. Verification Time
  - l. Verified By (e.g., "System", "Manual")
  - m. Comment (Comment added during manual verification by security team)
  - n. Drop Location (New - both summary and detail view)
5. Custom Fields (if configured): Example fields like Mode of Transport, Vehicle Details, Escort Required, Escort Name, Remarks, Commute by Ola/Metro/Bus/Auto.
6. Removals/Renames of fields which weren't relevant for WIS use case:
- a. Remove Shift Time.
  - b. Remove: Confirmed by THD, Confirmed by Co-Pax, Not Travelled, Confirmed by TPT.
  - c. Remove: Planned Sign-in Time and Actual Sign-in Time.
  - d. Rename Planned Drop Time to Planned Arrival Time.
  - e. Rename Actual Drop Time to Confirmed Arrival Time.
  - f. Remove 'P' from the visit time values in the report.
  - g. Remove Drop Address field from the detailed view panel.

### **5.3 System / Backend Flow (Supporting Use Cases)**

#### **Use Case 3.0: Phone Number Handling for Safe Reach Calls**

1. Problem: Safe Reach calls are triggered using an outdated phone number from the Employee DB, even if the employee provides an updated number during checkout.
2. Solution: The system will prioritize the latest phone number entered during checkout for Safe Reach calls.
3. Key Changes:
  - a. Phone Number Handling: The employee can edit their phone number during checkout. Safe Reach will trigger calls using the updated number provided during checkout. The updated number will be stored separately in the Safe Reach database and will not overwrite the Employee DB record.

- b. Safe Reach Workflow Integration: API logic will be modified to fetch the dynamically updated phone number for call triggering. The phone number used for Safe Reach calls will be validated and logged. Reports will reflect whether an employee's phone number was edited during checkout.

4. Edge Cases:

- a. Validation will prevent invalid phone number entries.
- b. If an employee doesn't edit their phone number, the existing workflow (using the DB number) remains unchanged.
- c. Only employee records are used when an employee is also a visitor.

Use Case 3.1: Configuring Safe Reach Time Window at Office Level

- 1. Problem: BUID-level configuration for Safe Reach start/end times limits flexibility for offices in different cities with varying safety protocols.
- 2. Solution: Introduce an office-level setting to define Safe Reach notification start and end times, which can override BUID-level settings.
- 3. Workflow Changes: If an Office-level Safe Reach time is configured, it will override the BUID-level time. If not configured at the Office-level, the system will fall back to existing BUID-level settings.

## Key Configurations/Properties

| Property Name                         | Description                                                                                                                            | Type                              |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Visitor Service Configurations</b> |                                                                                                                                        |                                   |
| SafeReachVmsTimeInMin                 | Defines the time window (in minutes from 00:00 AM) after which Safe Reach specific fields and workflow are triggered on the VMS kiosk. | Array; maintained at office-level |
| safeReachConsentContent               | Defines the consent statement content with a checkbox                                                                                  | Array; maintained at office-level |

|                                                     |                                                                                                                                          |                                     |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
|                                                     | displayed at the end of the Safe Reach form on the VMS kiosk.                                                                            |                                     |
| triggerSafeReachForFemaleOnly                       | Boolean flag to enable Safe Reach workflow and field visibility only for female employees.                                               | Boolean; maintained at BUID-level   |
| enableSafeReachForBookingTypes                      | Master boolean property that governs the overall functionality of Safe Reach. If false, triggerSafeReachForFemaleOnly cannot be enabled. | Boolean; maintained at BUID-level   |
| SafeReachInputFields                                | JSON structure to define custom input fields                                                                                             | JSON; maintained at office-level    |
| safeReachCcList                                     | List of email addresses to be CC'ed for failed Safe Reach verification emails.                                                           | Array; maintained at office-level   |
| enableSignatureForConsentSafeReach                  | Boolean flag to enable signature capture for NDA consent instead of a checkbox during Safe Reach checkout.                               | Boolean; maintained at office-level |
| etaToReachDestination (Within SafeReachInputFields) | Defines the configurable intervals for "ETA to Reach Destination" dropdown.                                                              | JSON; maintained at BUID-level      |
| enableSignatureForConsentSafeReach                  | To introduce signature for consent                                                                                                       | Boolean; maintained at office-level |
| KioskSafeReachInterval                              | To configure start and end time of safe reach trigger                                                                                    | Array; maintained at office-level   |
| <b>ETS-Side Configurations</b>                      |                                                                                                                                          |                                     |
| Mobile App Notification                             | Enable/disable mobile app notifications for Safe Reach.                                                                                  | Boolean                             |
| IVR Verification                                    | Enable/disable verification through IVR calls.                                                                                           | Boolean                             |

|                                                                         |                                                                                                                |                                     |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Gender                                                                  | Selection of gender(s) eligible for Safe Reach verification (if triggerSafeReachForFemaleOnly is false).       | Single-select                       |
| Start IVR at Sign off                                                   | Boolean to immediately trigger IVR on checkout.                                                                | Boolean                             |
| Mobile Notification Trigger time in mins                                | Default time after checkout to trigger mobile notification if ETA is not selected by employee.                 | Integer                             |
| IVR Trigger Buffer Time in mins                                         | Default time after which IVR should trigger if mobile notification is not set up and ETA is not provided.      | Integer                             |
| Time Difference Between Mobile App Notification And First IVR Call Time | Configurable time difference between mobile notification and the first IVR call.                               | Integer                             |
| Difference Between Two Consecutive IVR Calls                            | Configurable time difference between sequential IVR calls.                                                     | Integer                             |
| Drop verification failed email ID                                       | Default recipients for failed verification emails if safeReachCcList is not active for a specific BUID/Office. | List                                |
| <b>Employee Experience Internal Configurations</b>                      |                                                                                                                |                                     |
| wisFailDropEmailSubject                                                 | To define the subject line of failed verification email                                                        | String; maintained at BUID-level    |
| wisManualSuccessEmailSubject                                            | To define the subject line of manual successful email                                                          | String; maintained at BUID-level    |
| wisSafeReachReportingManagerEnabled                                     | Controls if reporting manager should be cc'ed or not in the failed verification email                          | Boolean; maintained at office-level |
| wisSafeReachCcList                                                      | Controls cc list of failed verification email                                                                  | Array; maintained at office-level   |

|                                                  |                                     |                                   |
|--------------------------------------------------|-------------------------------------|-----------------------------------|
| wisDropVerificationFailedEmail                   | To enable failed verification email | Boolean; maintained at BUID-level |
| wisDropVerificationManualSuccessEmail            | To enable manual successful email   | Boolean; maintained at BUID-level |
| <b>Project Management Service Configurations</b> |                                     |                                   |
| SAFE_REACH_ENABLED                               | To enable safe reach dashboard      | Boolean; maintained at BUID-level |

## Dependencies/Integrations

1. VMS Kiosk: Primary interface for employee checkout and Safe Reach initiation.
2. Employee Database: Source of truth for employee details, prefilling information, and gender checks.
3. Safe Reach Module (Core): Handles the logic for triggering notifications, status tracking, and data storage.
4. WIS-PMS (Property Management System): Manages all configurable properties for Safe Reach.
5. IVR System: For automated voice calls for verification.
6. Mobile Notification System: For sending push notifications to employees on their app.
7. Email Service: For sending various email alerts and confirmations.
8. Database: For storing all Safe Reach related data, including audit trails and custom fields.

## Notifications

Multiple notification types are integral to the Safe Reach workflow:

1. Success Message (VMS Kiosk):
  - a. Content: "Safe reach has been initiated. You will receive a call in [Selected ETA or Configured Mobile Notification Trigger Time] mins to confirm if you

have reached your destination. Please answer the call promptly for your safety."

- b. Trigger: Immediately upon successful employee checkout through VMS.

2. Mobile App Notification:

- a. Trigger: After the "ETA to Destination" time selected by the employee, or after the configured mobile notification trigger time if ETA is not selected.
- b. Content: Prompts employee to confirm safe arrival.
- c. Response: Employee can respond via the app.

3. IVR Calls:

- a. Trigger: Triggered sequentially if mobile notification fails or is not responded to, or as a default if mobile notification/ETA is not configured.
- b. Number of Calls: Up to 3 IVR calls.
- c. Content: "Press 1 to confirm you have reached, Press 2 if you are still traveling."

4. Failed Verification Emails:

- a. Recipient: Employee, and relevant Office Admins (based on configured CC lists or fallbacks).
- b. Trigger: At configurable intervals (e.g., every 30 minutes) if automated (mobile/IVR) verification attempts fail and manual verification is not completed.